STT 180 Group 1 Presentation

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Credit dataset from "An Introduction to Statistical Learning(ISL)" by James, Witten, Hastie and Tibshirani

Which variables are most strongly associated with credit card limits?

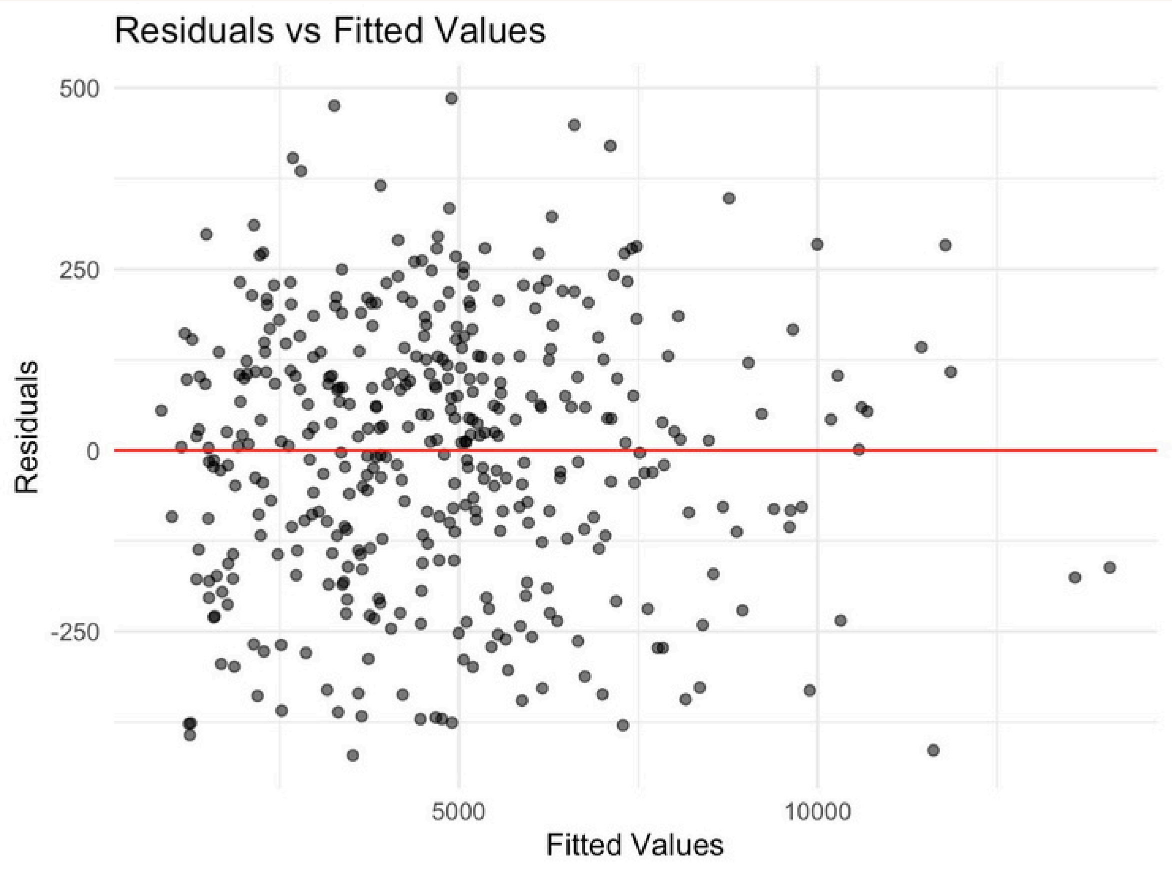
What relationships exist between students and credit card data?

Introduction

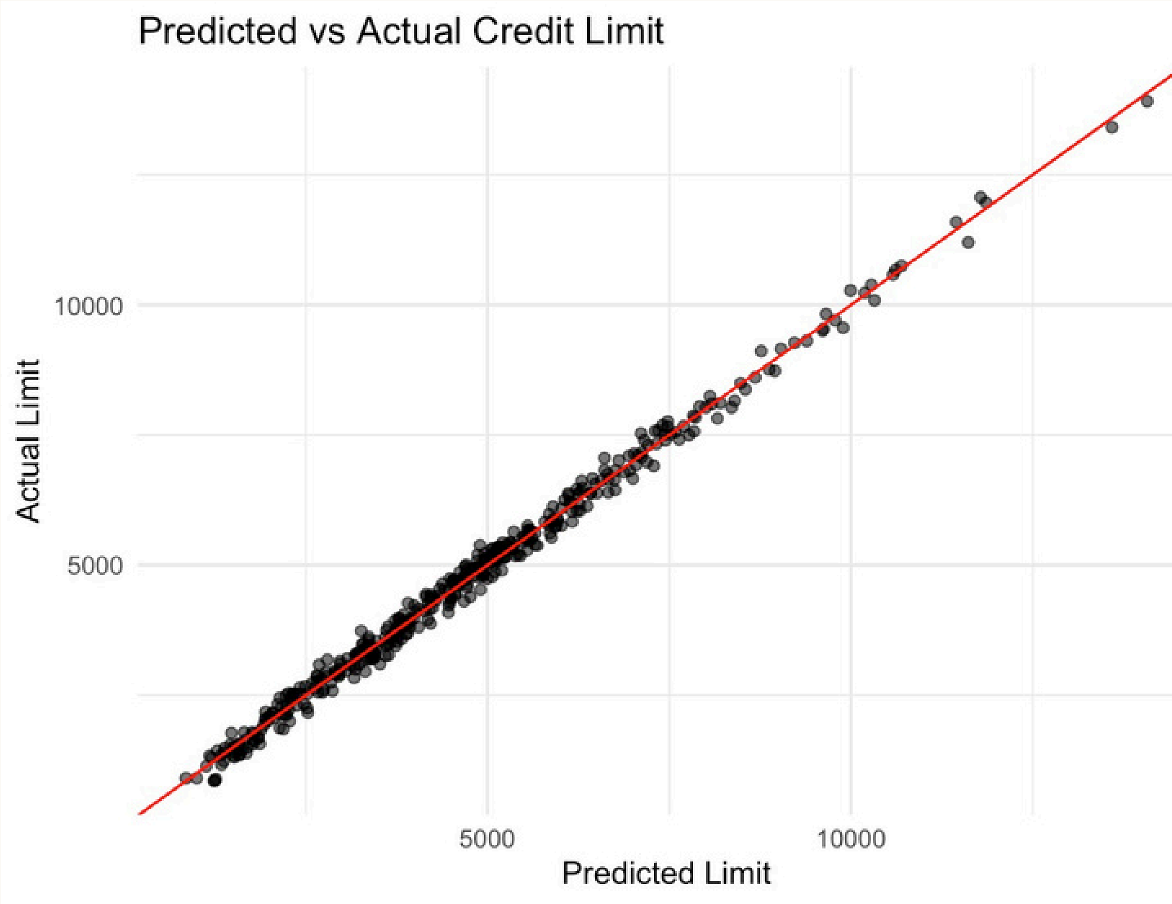
What is the correlation between a person's credit score and their marital status?

Financial literacy is often overlooked or not fully understood, especially by students. So, the primary aim of this study is to determine which factors have the greatest impact on credit card limits and to examine patterns in credit behavior among students.

What factors have the greatest influence on an individual's credit limit?



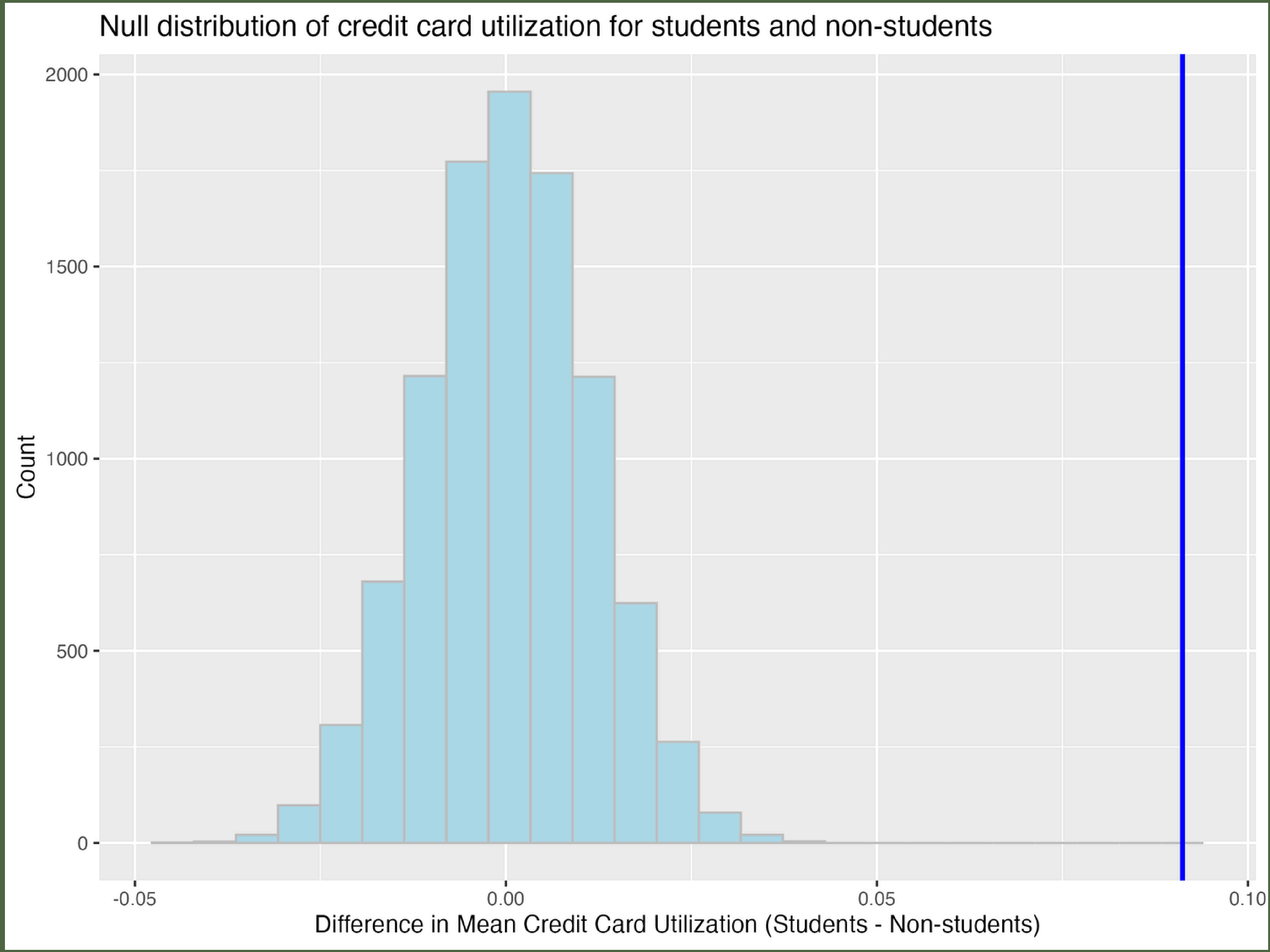
1>Created ggplots for different variables v/s the limit, found their correlation to see which variable is more strongly associated
2>Built a multiple regression model predicting Credit Limit using Rating and Income. Used the p-values and coefficients to comprehend the variables' contributions.
3>Generated Predicted vs Actual Credit Limit and Residual vs Fitted to assess model fit



$$L = -532.47 + 14.77R + 0.56I$$

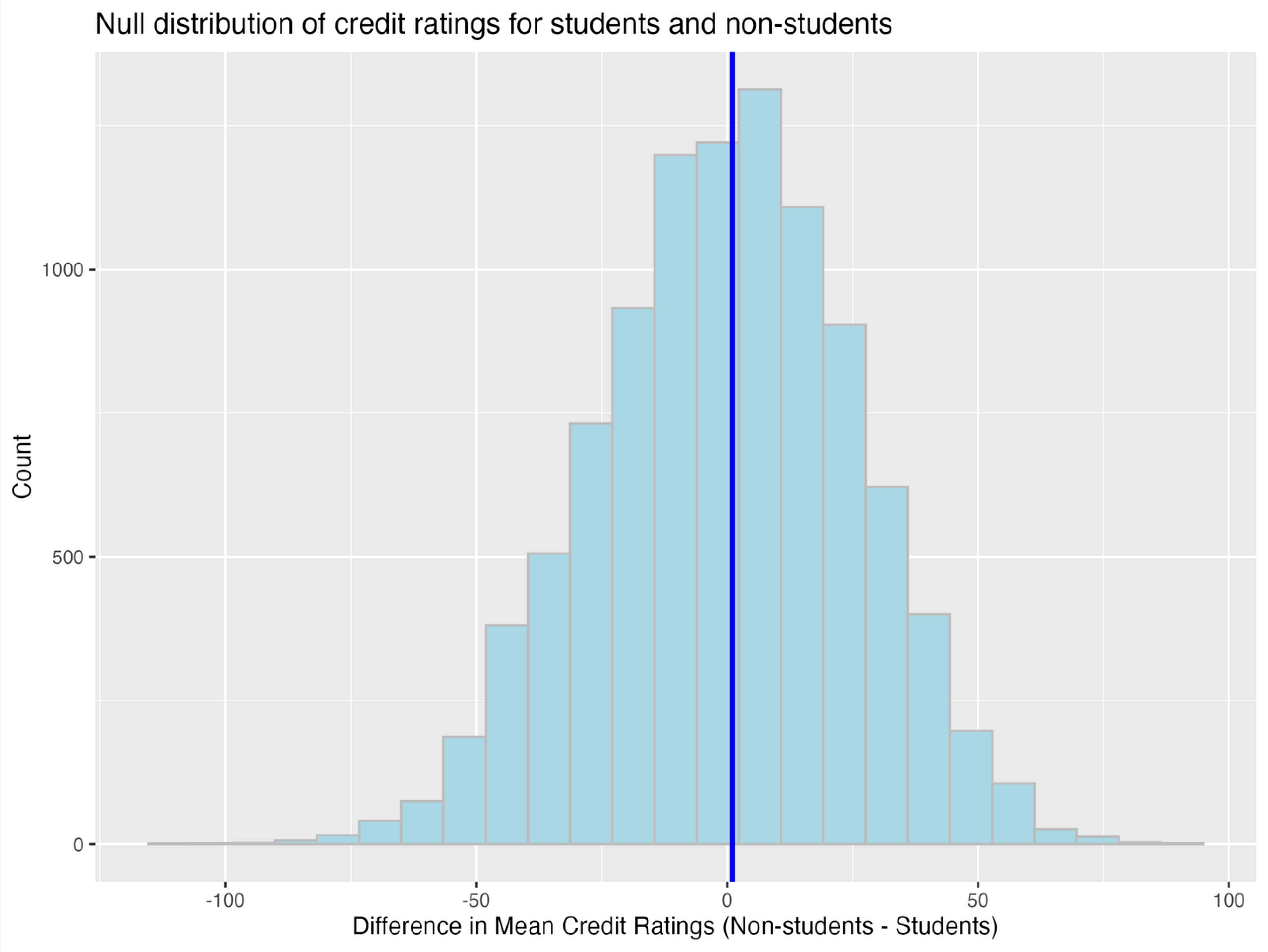
Do students have higher utilization than non-students?

- Created null distribution
  - 10,000 bootstrap sample differences in means
- Calculated sample statistic
  - Observed difference in mean utilization (students - nonstudents)
  - Observed difference = 0.0912
- Calculated p-value
  - p = 0
- Reject null, evidence for student utilization being higher than nonstudent



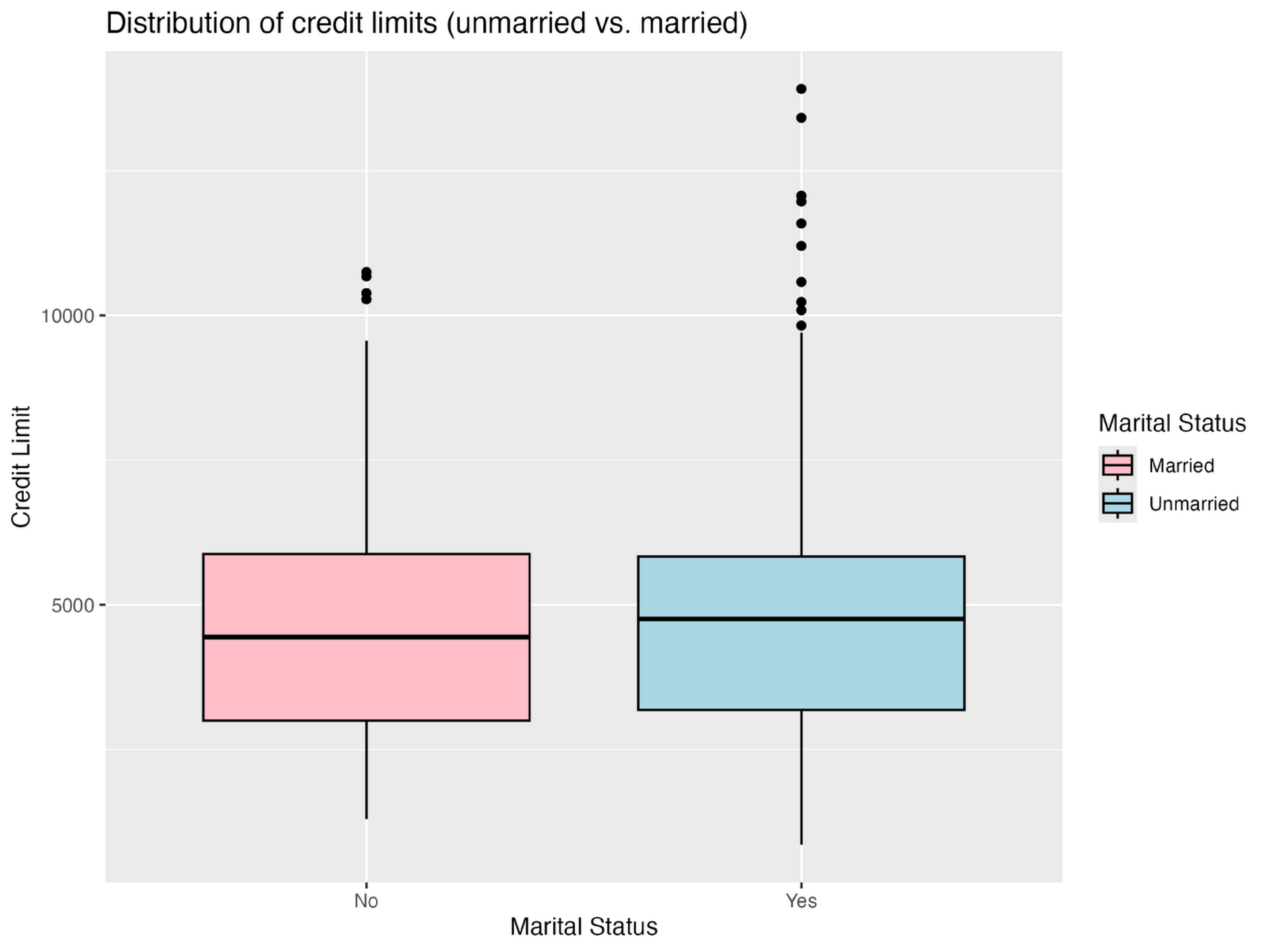
Do non-students have higher credit scores than students?

- Created null distribution
  - 10,000 bootstrap sample differences in means
- Calculated sample statistic
  - Observed difference in mean credit scores (nonstudents - students)
- Calculated p-value
  - p = 0.4834
- Fail to reject the null hypothesis, no convincing evidence that non-students have higher credit scores than students



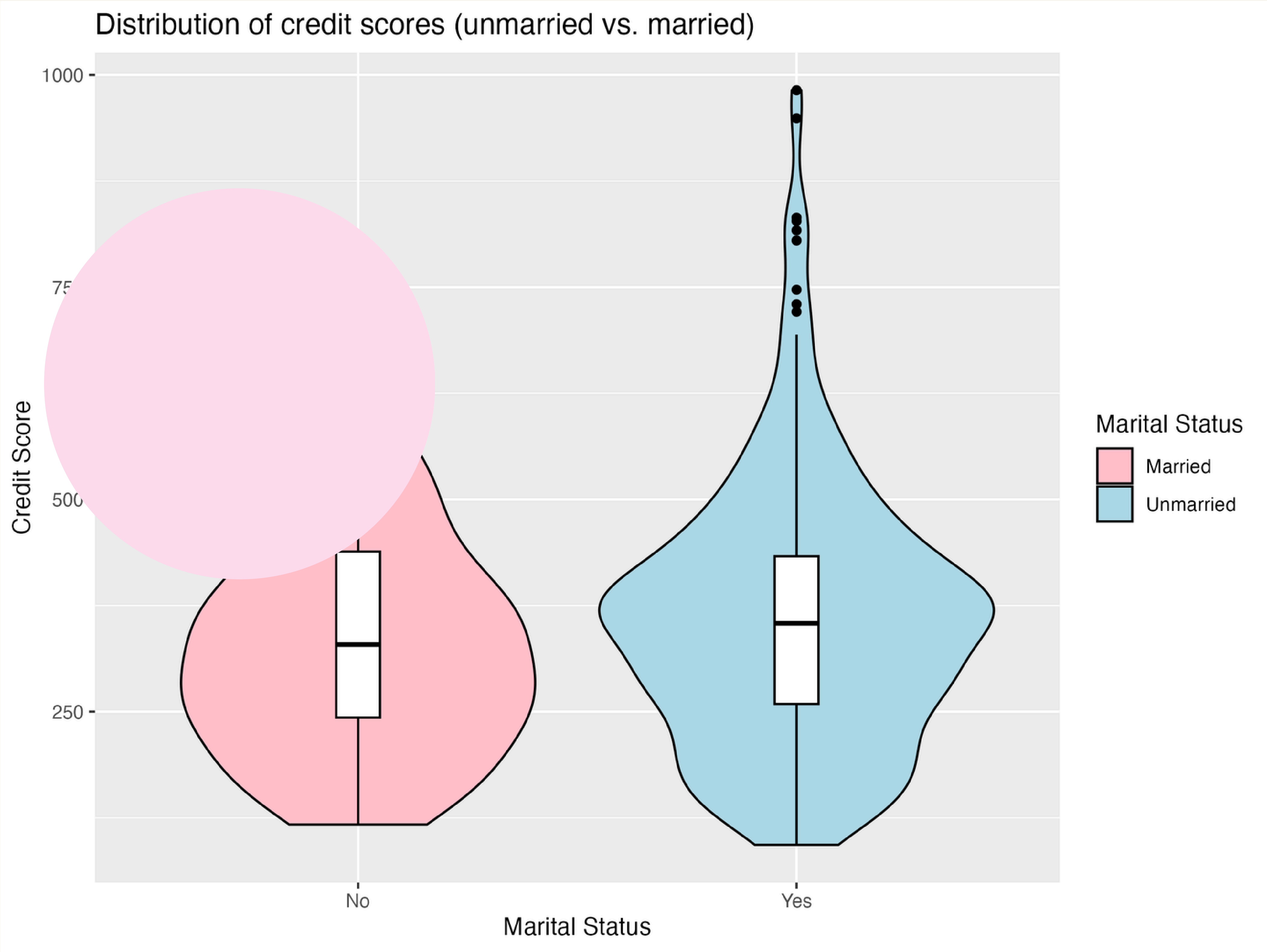
Do married individuals have a higher credit limit than unmarried individuals?

- Plotted distribution of credit limit for married and unmarried people to find initial difference
- Created a bootstrap distribution for difference in mean credit limit (married - unmarried)
  - 10,000 bootstrap samples
- Calculated 95% confidence interval for true difference in mean
  - Interval = -304.1433 to 531.4101
- 0 is contained in interval, results not significant



Is there a relationship between marital status and credit score?

- Plotted distribution of credit scores for married and unmarried people to find initial difference
- Conducted Chi-square test for independence for these variables
  - Used chisq.test()
- Calculated p-value
  - p = 0.3869
- Fail to reject null hypothesis, no convincing evidence of a relationship between marital status and credit score



Conclusion

What is the relationship between credit score and available loan amounts?

~Credit rating and income had a significant impact on credit limit
~There was no significant evidence to support that marriage status had an impact on credit score.
~Students tended to have higher credit utilizations than non-students.
~No significant difference in credit rating between students and non-students.

How does debt levels affect credit score?

What is the relationship between credit score and available mortgage rates?

~Limited by our knowledge and proficiency with both statistics and the use of the programming language R.
~ The data set was limited to the variables they provided. Further, the origins and description of the variables of our data set was somewhat limited.

References

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